Report: Advertising Sales Prediction Model Evaluation

1. Data Registration

The dataset used in this project focuses on predicting sales based on advertising data. After loading the data, the following preprocessing steps were performed:

- **Missing Values**: No missing values were identified in the primary columns used for analysis.
- **Outlier Handling**: Outliers were clipped to the 5th and 95th percentiles to ensure robust model training.
- **Feature Splitting**:
  - Features (X) included all columns except the target column sales.
  - Target (y) was set to the sales column.

2. Model Evaluation Metrics

The following metrics were computed for each model:

- **Mean Absolute Error (MAE)**: Measures the average magnitude of errors in predictions.
- **Mean Squared Error (MSE)**: Measures the squared differences between predicted and actual values.
- **R-squared (R^2)**: Indicates how well the model explains the variance in the target variable.

MODEL NAME	MAE	MSE	R2
GRADIENT BOOSTING (V1)	2.0193	5.3945	0.8191
LINEAR REGRESSION	1.4309	2.9819	0.9000
K-NEAREST NEIGHBORS (KNN)	1.1331	2.4235	0.9187
GRADIENT BOOSTING (V2)	0.5623	0.4920	0.9835
RANDOM FOREST	0.5722	0.4852	0.9837

- gradient_boosting_model
- linear_regression_model
- knn_model
- gradient_boosting_model_v2
- random_forest_model

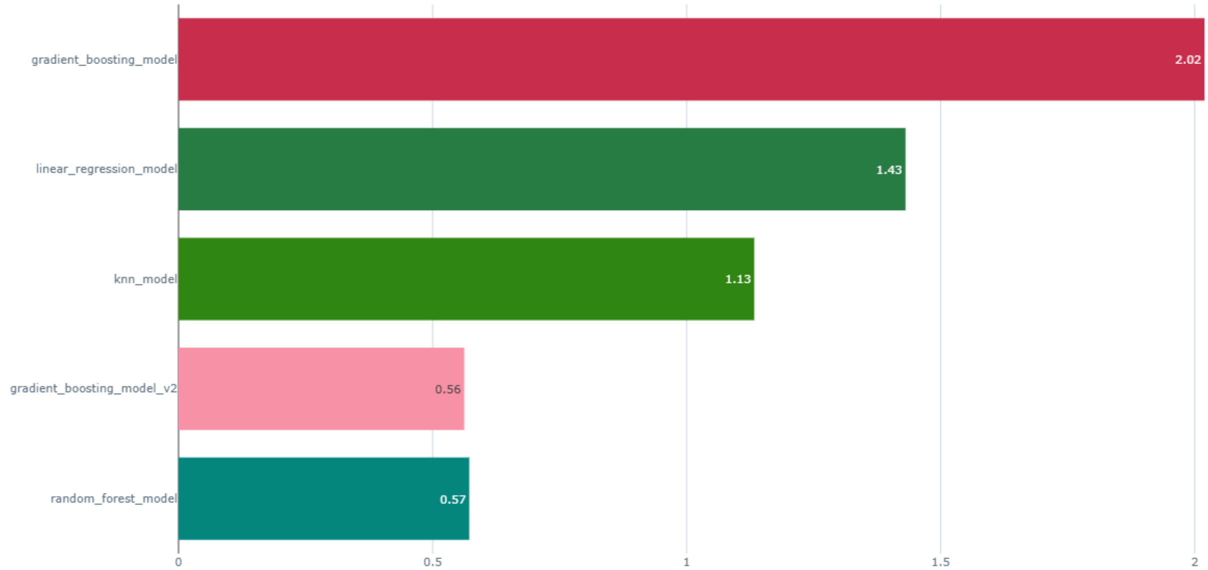


Figure 1:MAE

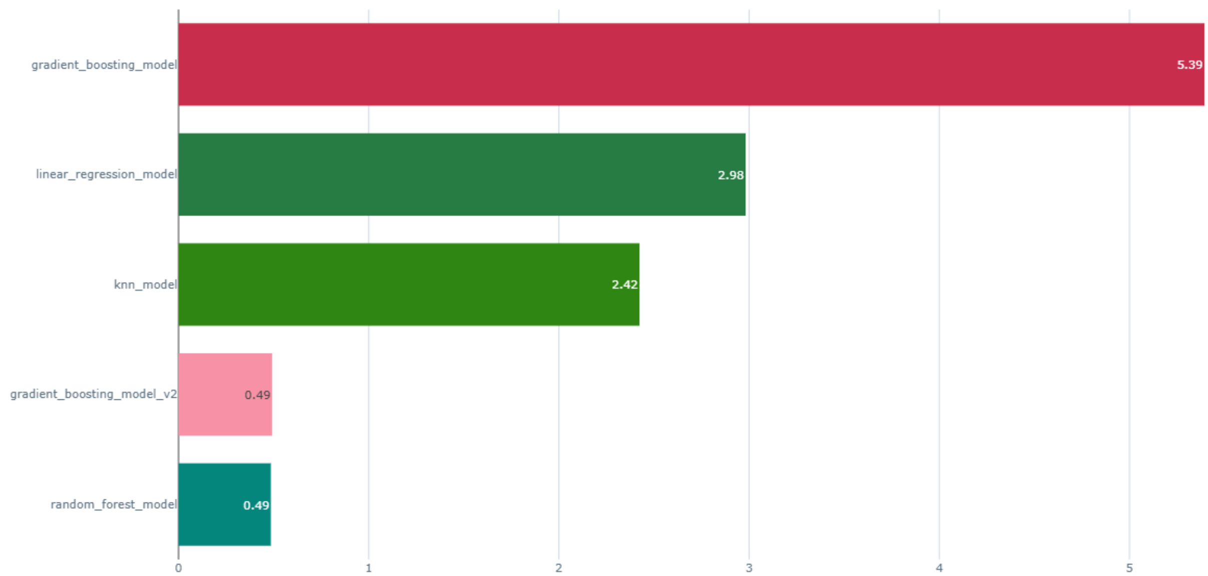


Figure 2:MSE

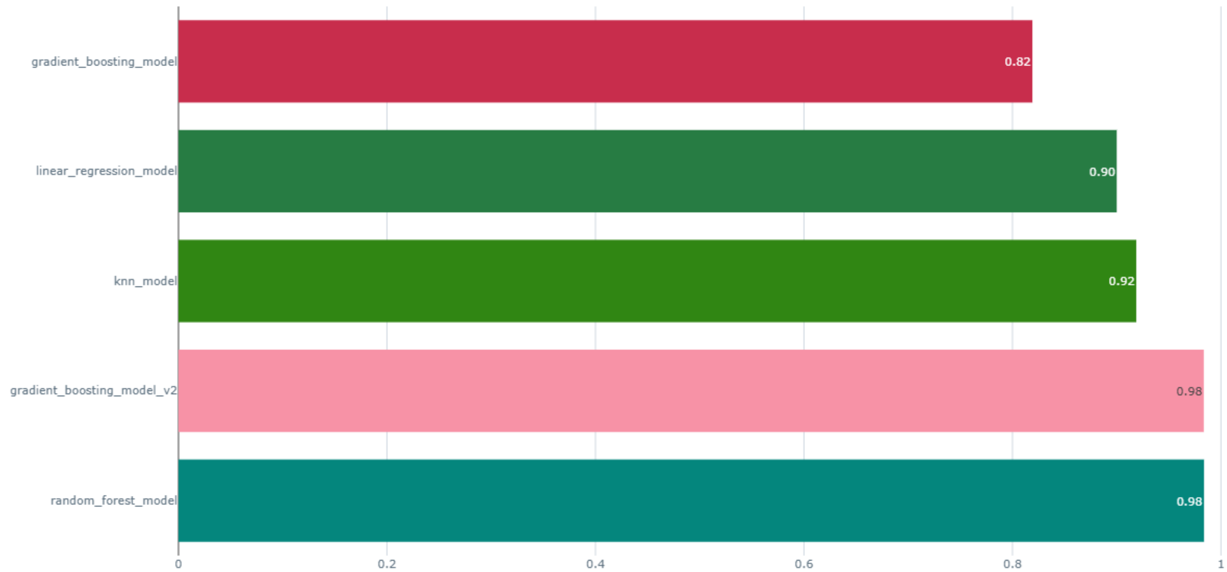


Figure 3: R^2

3. Model Comparison

Key Observations:

1. **Gradient Boosting (v1):**
   - Had the lowest performance among the models tested, with an R^2 of 0.8191.
   - MSE was relatively high at 5.3945, indicating room for improvement in prediction accuracy.
2. **KNN:**
   - Performed well with an R^2 of 0.9187 but fell short compared to Gradient Boosting (v2) and Random Forest.
   - MAE and MSE were higher than the top-performing models.
3. **Linear Regression:**
   - Achieved decent performance with an R^2 of 0.9000.
   - Suitable for tasks prioritizing interpretability despite lower accuracy compared to tree-based methods.
4. **Gradient Boosting (v2):**
   - Performed exceptionally well, achieving an R^2 of 0.9835.
   - Lowest MAE (0.5623) and second-lowest MSE (0.4920) among all models.

5. **Random Forest:**
   - Achieved the best overall performance with an R^2 of 0.9837.
   - MSE (0.4852) and MAE (0.5722) were comparable to Gradient Boosting (v2), making it a strong candidate for deployment.

4. Offline Evaluations

Offline evaluation was conducted by splitting the dataset into training (80%) and validation (20%) subsets. The performance of each model on the validation set was logged using MLflow, ensuring traceability and reproducibility.

Data Logging :

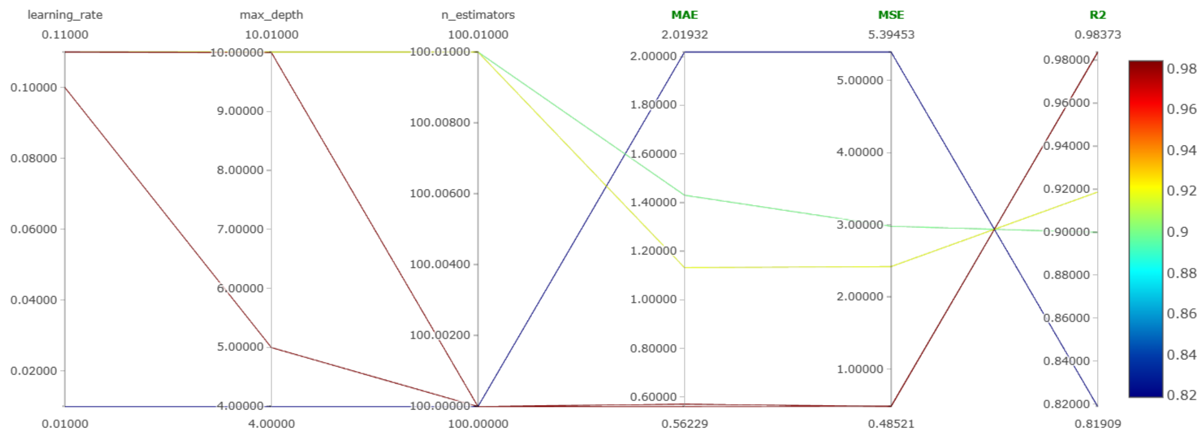
Data logging is the foundational step in tracking machine learning experiments. In the presented MLflow interface, critical data parameters were logged to ensure reproducibility and detailed experimentation.

Logged Parameters

PARAMETER	GRADIENT BOOSTING (V1)	KNN	LINEAR REGRESSION	GRADIENT BOOSTING (V2)	RANDOM FOREST
ALGORITHM	-	auto	-	-	-
FIT_INTERCEPT	-	-	True	-	-
LEARNING_RATE	0.01	-	-	0.1	-
MAX_DEPTH	4	-	-	5	10
MIN_SAMPLES_LEAF	-	-	-	-	1
MIN_SAMPLES_SPLIT	-	-	-	-	2
N_ESTIMATORS	100	-	-	100	100
N_NEIGHBORS	-	4	-	-	-
RANDOM_STATE	42	-	-	42	42
WEIGHTS	-	uniform	-	-	-

Performance and Metrics Tracking

METRIC	GRADIENT BOOSTING (V1)	KNN	LINEAR REGRESSION	GRADIENT BOOSTING (V2)	RANDOM FOREST
DURATION (S)	4.5	4.3	4.3	4.2	8.4
MAE	2.0193	1.1331	1.4309	0.5623	0.5722
MSE	5.3945	2.4235	2.9819	0.4920	0.4852
R^2	0.8191	0.9187	0.9000	0.9835	0.9837



5. Conclusion

- **Gradient Boosting (v2)** and **Random Forest** are the top-performing models, excelling in accuracy and error metrics.
- For tasks prioritizing interpretability, Linear Regression remains a viable option.
- **KNN** may require further parameter tuning to enhance its performance.

Recommendations:

1. Experiment with hyperparameter tuning for Random Forest and Gradient Boosting to explore potential improvements.
2. Consider adding more features or transforming existing ones to enhance model accuracy.
3. Use cross-validation for a more robust evaluation of model performance.
4. Perform hyperparameter optimization on Gradient Boosting (v2) and Random Forest to explore further improvements.
5. Consider cross-validation for more robust evaluation of model performance.Evaluate additional feature engineering or transformation techniques to improve model accuracy.

Appendix

MLflow Experiment Tracking: All model runs were logged to the local MLflow server. The tracking server provides a comprehensive record of parameters, metrics, and artifacts for future reference and analysis.